DA5000W Linear algebra and optimization

Assignment 2

October 2025

Topics:

1. Linear Transformations
2. Properties of Linear Transformations
3. Eigen value and Eigen vector
4. Matrix factorization
5. SVD

Question 1:
For the following set of problems,

- Prove that T is a linear transformation.
- Find bases for both $N(T)$ and $R(T)$.
- Compute the nullity and rank of T.
- Verify the dimension theorem.
- Determine whether T is one-to-one or onto.

(a) $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$T(a_1, a_2, a_3) = (a_1 - a_2,\ 2a_3).$$

(b) $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$T(a_1, a_2) = (a_1 + a_2,\ 0,\ 2a_1 - a_2).$$

(c) $T : M_{2\times3}(F) \to M_{2\times2}(F)$ defined by

$$T\left(\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}\right) = \begin{bmatrix} 2a_{11} - a_{12} & a_{13} + 2a_{12} \\ 0 & 0 \end{bmatrix}.$$

(d) $T : P_2(\mathbb{R}) \to P_3(\mathbb{R})$ defined by

$$T(f(x)) = xf(x) + f'(x).$$

Question 2:
Given a set of linear equations

$$x + y = 2, \quad x + 2y = 3, \quad x + 3y = 4.$$

1. Write in matrix form
2. Find the basis for column space and null space
3. Interpret these subspaces geometrically

Question 3:
In $\mathbb{R}^2$, let L be the line $y = mx$, where $m \neq 0$.
Find an expression for $T(x, y)$, where T is the reflection of $\mathbb{R}^2$ about L.
Hint: Reflect by rotating the line to the x-axis, apply $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, then rotate back.

$$T = R(\theta)\,\mathrm{diag}(1, -1)R(-\theta) \quad \text{with} \quad \theta = \arctan m.$$

Question 4:
For each of the following matrices $A \in M_{n\times n}(\mathbb{R})$

1. test A for diagonalizability
2. find an invertible matrix Q and a diagonal matrix D such that $Q^{-1}AQ = D$.

$$A_1 = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}, A_3 = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{bmatrix}$$

Note: You have to perform the calculation for A_1, A_2 and A_3

Question 5:
(a) A 2×2 symmetric matrix has eigenvectors v1 and v2, corresponding to two distinct eigenvalues.
Find the value of a.

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 8 \\ -a \end{bmatrix},$$

(b) You are given a 3×3 matrix A with the following properties:

- $\mathrm{trace}(A) = 4$
- $\det(A) = -18$
- One of its eigenvalues is $\lambda_1 = -2$.

You are also told how A acts on a specific vector $\mathbf{v}$:

$$A \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix}$$

Using only this information, answer the following questions:

1. What are the other two eigenvalues of A?
2. What is the result of $A^5\mathbf{v}$? (where $\mathbf{v}$ is the vector given above)

3. Is the matrix $(A - 3I)$ invertible? Justify your answer.

Question 6:
An agricultural model tracks the yearly populations of three interacting species: Wheat (W), Weevil (V), and a specialist Fungus (F). The population vector is $\mathbf{x}_k = \begin{bmatrix} W_k \\ V_k \\ F_k \end{bmatrix}$, and its evolution is governed by $\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$, where the interaction matrix $\mathbf{A}$ is given by:

$$\mathbf{A} = \begin{bmatrix} 4 & 0 & 0 \\ 1 & 3 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

Your task is to analyze the long-term dynamics of this system:

1. Find the eigenvalues (λ) of the matrix $\mathbf{A}$.
2. Find the corresponding eigenvectors $(\mathbf{v})$ for each eigenvalue.
3. Follow-up: The model matrix $\mathbf{A}$ is diagonalizable. Explain the significance of the matrix $\mathbf{P}$ in the diagonalization $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, and what this entire factorization reveals about predicting the population $\mathbf{x}_k$ after many years $(k \to \infty)$.

Question 7:
A grayscale image is represented as a 512×512 matrix A, where each entry a_{ij} represents the pixel intensity at position (i, j). The singular value decomposition of A is given by $A = U\Sigma V^T$, where the first few singular values are: $\sigma_1 = 15000$, $\sigma_2 = 8000$, $\sigma_3 = 3000$, $\sigma_4 = 500$, $\sigma_5 = 200$, and $\sigma_i < 50$ for all $i > 5$.

(a) Explain the concept of low-rank approximation using SVD for image compression. What information do the singular values represent about the image?

(b) The rank-k approximation of A is defined as $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T$. Calculate the storage requirement (number of values to store) for the original matrix A versus the rank-3 approximation A_3. What is the compression ratio?

Question 8: Let

$$A_1 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, A_2 = \begin{bmatrix} -2 & 8 & 20 \\ 14 & 19 & 10 \\ 2 & -2 & 1 \end{bmatrix}$$

1. Compute $A^T A$ and AA^T.
2. Find their eigenvalues and unit eigenvectors.
3. Construct the singular value decomposition $A = U\Sigma V^T$.
4. Multiply $U\Sigma V^T$ to recover A.

Note: You have to perform the calculation for both A_1 and A_2

Question 9:
Consider the symmetric matrix

$$S = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

1. Compute the eigenvalues and eigenvectors of S.
2. Verify that the eigenvectors are orthogonal and form a unitary matrix Q.
3. Perform SVD

Coding Assignment
Question 10:
Application of SVD for image noise reduction.

Using the provided noisy grayscale image (Link), apply Singular Value Decomposition (SVD) to perform image noise reduction by reconstructing the image using a lower-rank approximation. Plot and analyze how image quality changes as you vary the number of singular values retained.

1. Plot the singular values vs rank.
2. Decide an appropriate rank cutoff for reconstruction and justify your choice based on the singular value distribution and image quality.
3. Reconstruct and display the image for different ranks (e.g., r = 5, 10,... and your chosen cutoff rank).

Code submission guidelines

1. Prepare your assignment exclusively in a Jupyter Notebook (.ipynb format).
2. Ensure all code cells are executed and their corresponding outputs are visible.
3. Name the notebook as following: $\{Rollnumber\}\_LA\_MFDS\_Assignment\_2.ipynb$